

Part Name/Description	Unit Quantity	Unit Prototype Cost	Total Prototype Cost	Unit Production Cost
ESP 32 - Wroom N8	1	\$5.49	\$5.49	\$5.49
CAP CER 330PF 16V X7R	2	\$0.10	\$0.20	\$0.10
CAP CER 10000PF 10V X	2	\$0.08	\$0.16	\$0.08
CAP CER 0.022UF 6.3V X	1	\$0.10	\$0.10	\$0.10
CAP 0805 250V 33 NF X7	6	\$0.18	\$1.08	\$0.18
CAP CER 0.1UF 10V X5R	7	\$0.08	\$0.56	\$0.08
CAP CER 0.47UF 6.3V X5	1	\$0.08	\$0.08	\$0.08
CAP CER 2.2UF 16V X5R	2	\$0.10	\$0.20	\$0.10
CAP CER 10UF 16V X5R	1	\$0.08	\$0.08	\$0.08
CAP CER 22UF 6.3V 20%	4	\$0.28	\$1.12	\$0.28
CAP CER 100UF 16V X5R	5	\$1.74	\$8.70	\$1.74
CAP CER 150UF 6.3V X5R	1	\$1.31	\$1.31	\$1.31
RES 10 OHM 5% 2W 2512	1	\$0.44	\$0.44	\$0.44
RES 20 OHM 5% 2W 2512	1	\$0.24	\$0.24	\$0.24
RES 220 OHM 5% 2W 251	1	\$0.53	\$0.53	\$0.53
RES 1.65K OHM 1% 2W 1	1	\$1.39	\$1.39	\$1.39
RES 5.11K OHM 1% 2W 2	1	\$0.52	\$0.52	\$0.52
RES 10K OHM 1% 2W 25	1	\$0.51	\$0.51	\$0.51
RES 49.9K OHM 1% 2W 2	1	\$0.52	\$0.52	\$0.52
DIODE SCHOTTKY 100V	5	\$0.40	\$2.00	\$0.40
LED GREEN CLEAR SMD	1	\$0.10	\$0.10	\$0.10
FUSE GLASS 5A 250VAC	1	\$0.63	\$0.63	\$0.63
FUSE GLASS 4A 250VAC	1	\$0.63	\$0.63	\$0.63
FIXED IND 4.7UH 2.5A 14	1	\$1.03	\$1.03	\$1.03
FIXED IND 33UH 2.5A 93.	1	\$1.06	\$1.06	\$1.06
Positional Rotation DC Mo	2	\$5.95	\$11.90	\$5.95
STANDARD MOTOR 1160	3	\$3.71	\$11.13	\$3.71
8 Pin Connector	2	\$1.48	\$2.96	\$1.48
8 Pin Ribbon Cable	2	\$5.00	\$10.00	\$5.00
Power Barrel Connector Ja	1	\$0.71	\$0.71	\$0.71
USB - micro B USB 2.0	1	\$0.77	\$0.77	\$0.77
Switch button	2	\$0.10	\$0.20	\$0.10
SWITCH PB DPDT 0.1A 3	1	\$0.46	\$0.46	\$0.46
IC HALF BRIDGE DRIVER	3	\$3.55	\$10.65	\$3.55
Buck Switching Regulator	1	\$3.77	\$3.77	\$3.77
Buck Switching Regulator	1	\$0.71	\$0.71	\$0.71
3 Pin Screw Terminal	2	\$0.71	\$1.42	\$0.71
2 Pin Screw Terminal	3	\$0.32	\$0.96	\$0.32
		Total Prototype cost	\$78.63	Total Production cost

Bill of Materials Motor Subsystem

Total Production Cost	Manufacturer	Manufacturer Part #	Vendor Link	Datasheet Link
\$5.49	Espressif	ESP32-S3-WROOM-1-	https://www.digikey.com	https://documentation.espressif.com
\$0.20	Murata	GRM033R71C331KA0	https://www.digikey.com	https://search.murata.com
\$0.16	Murata	GRM022C71A103KE0	https://www.digikey.com	https://search.murata.com
\$0.10	Murata	GRM033R60J223KE01	https://www.digikey.com	https://search.murata.com
\$1.08	Yageo	CC0805KKX7RYBB33	https://www.digikey.com	https://www.yageo.com
\$0.56	Murata	GRM033R61A104KE15	https://www.digikey.com	https://search.murata.com
\$0.08	Murata	GRM033R60J474KE90	https://www.digikey.com	https://search.murata.com
\$0.20	Samsung Electr	CL10A225KO8NNNC	https://www.digikey.com	https://mm.digikey.com
\$0.08	Taiyo Yuden	EMK212BJ106KG-T	https://www.digikey.com	https://mm.digikey.com
\$1.12	Murata	GRM155R60J226ME1	https://www.digikey.com	https://search.murata.com
\$8.70	Taiyo Yuden	EMK325ABJ107MM-T	https://www.digikey.com	https://mm.digikey.com
\$1.31	Samsung Electr	CL32A157MQVNNNE	https://www.digikey.com	https://weblib.samsung.com
\$0.44	Bourns	CRM2512-JW-100ELF	https://www.digikey.com	https://www.bourns.com
\$0.24	Yageo	AC2512JK-7W20RL	https://www.digikey.com	https://yageogroup.com
\$0.53	TE Connectivity	CRGH2512J220R	https://www.digikey.com	https://www.te.com/
\$1.39	TE Connectivity	3503G2B1K65FTD	https://www.digikey.com	https://www.te.com/
\$0.52	Bourns	CRM2512-FX-5111ELF	https://www.digikey.com	https://www.bourns.com
\$0.51	TE Connectivity	CRGP2512F10K	https://www.digikey.com	https://www.te.com/
\$0.52	Bourns	CRM2512-FX-4992ELF	https://www.digikey.com	https://www.bourns.com
\$2.00	Vishay	VS-30WQ10FNTR-M3	https://www.digikey.com	https://www.vishay.com
\$0.10	Lite-On	LTST-C170GKT	https://www.digikey.com	https://mm.digikey.com
\$0.63	Schurter	34.3124	https://www.digikey.com	https://www.digikey.com
\$0.63	Schurter	34.3123	https://www.digikey.com	https://www.schurter.com
\$1.03	TDK	SPM4020T-4R7M-LR	https://www.digikey.com	https://product.tdk.com
\$1.06	Murata	1274AS-H-330M=P3	https://www.digikey.com	https://search.murata.com
\$11.90	Adafruit	4326	https://www.digikey.com	https://mm.digikey.com
\$11.13	NMB	PPN7PA12C1	https://www.digikey.com	https://mm.digikey.com
\$2.96	Molex	702460801	https://www.digikey.com	https://www.molex.com
\$10.00	Amphenol	10124667-Z0025YYLF	https://www.digikey.com	https://cdn.amphenol.com
\$0.71	Same Sky	PJ-102AH	https://www.digikey.com	https://www.cuidevice.com
\$0.77	GCT	USB3131-30-0230-A	https://www.digikey.com	https://gct.co/files/st
\$0.20	Same Sky	TS02-66-55-BK-100-LC	https://www.digikey.com	https://www.samesky.com
\$0.46	E-Switch	TL2233OA	https://www.digikey.com	https://configured-pi
\$10.65	Infineon	IFX9201SGAUMA1	https://www.digikey.com	https://www.infineon.com
\$3.77	Texas Instrume	LM5005MHX/NOPB	https://www.digikey.com	https://www.ti.com/c
\$0.71	Diode Incorpora	AP63203WU-7	https://www.digikey.com	https://www.diodes.com
\$1.42	Adam Tech	EBWA-03-A	https://www.digikey.com	https://app.adam-te
\$0.96	Same Sky	TB002-500-02BE	https://www.digikey.com	https://www.cuidevice.com
\$78.63				

Supplier	Supplier Part	# Ordered	Date Ordered	# Received	Surplus	Schematic Reference Designators
Digikey	5407-ESP32-S3-WROOM-1-N8CT-ND				-1	U4
Digikey	490-11359-1-ND				-2	C5,C10
Digikey	490-GRM022C71A103KE01LCT-ND				-2	C1,C7
Digikey	490-3169-1-ND				-1	C8
Digikey	13-CC0805KKX7RYBB333CT-ND				-6	C23,C24,C25,C26,C27,C28
Digikey	490-5881-1-ND				-7	C11,C16,C18,C21,C22,C30,C32
Digikey	490-10409-1-ND				-1	C4
Digikey	1276-1040-1-ND				-2	C2,C3
Digikey	587-1295-1-ND				-1	C6
Digikey	490-GRM155R60J226ME11DCT-ND				-4	C9,C12,C14,C15
Digikey	587-3152-1-ND				-5	C17,C19,C20,C29,C31
Digikey	1276-3367-1-ND				-1	C13
Digikey	CRM2512-JW-100ELFCT-ND				-1	R5
Digikey	13-AC2512JK-7W20RLCT-ND				-1	R2
Digikey	A126460CT-ND				-1	R9
Digikey	1712-3503G2B1K65FTDCT-ND				-1	R3
Digikey	118-CRM2512-FX-5111ELFCT-ND				-1	R4
Digikey	A130733CT-ND				-1	R6,R7,R10,R11
Digikey	118-CRM2512-FX-4992ELFCT-ND				-1	R1
Digikey	112-VS-30WQ10FNTR-M3CT-ND				-5	D1,D2,D3,D4,D6
Digikey	160-1179-1-ND				-1	D5
Digikey	486-1223-ND				-1	F1
Digikey	486-1222-ND				-1	F2
Digikey	445-174477-1-ND				-1	L2
Digikey	490-10851-1-ND				-1	L1
Digikey	1528-4326-ND				-2	M1,M2
Digikey	942-PPN7PA12C1-ND				-3	M3, M4, M5
Digikey	900-0702460801-ND				-2	J3, J4
Digikey	10124667-Z0025YYLF-ND				-2	J3, J4
Digikey	CP-102AH-ND				-1	J1
Digikey	2073-USB3131-30-0230-ACT-ND				-1	J2
Digikey	2223-TS02-66-55-BK-100-LCR-D-ND				-2	SW1,SW2
Digikey	141-TL2233OACT-ND				-1	S1
Digikey	IFX9201SGAUMA1CT-ND				-3	U1,U2,U3
Digikey	LM5005MHX/NOPBCT-ND				-1	VR1
Digikey	AP63203WU-7DICT-ND				-1	VR2
Digikey	2057-EBWA-03-A-ND				-2	M1,M2
Digikey	102-6145-ND				-3	M3, M4, M5